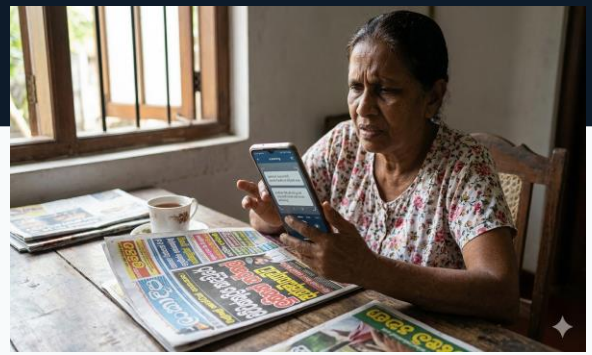
Interactive Sinhala Text Reader for the Visually Impaired with the assistance of OCR, NLP, and AI.

Group R26-DS-002 | Sri Lankan Institute of Information Technology

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Student ID	Comments	How these are addressed
IT22072238	• CER is a bit high • Using tesseract as a base comparison was good but rather sample level comparison run it for a test dataset for CER comparisons • Topic Identification part has not been completed	• Changed from a Res-net model to tesseract which makes use of LSTM to read word wise instead of character wise • The same Tesseract model was trained on two different version of the same dataset, with one have MAT skeletonization, and other having original lettering. • The main focus is simplifying complex character to a skeletonized form.
IT22004468	• What is the use of the adaptive TTS model • How is it adaptive • Are the two LLM's fine tuned	• Not a TTS model - it is the Adaptive Personalization Layer • Online learning (River Softmax Regression) that learns incrementally per turn, per user, with no pre-trained dataset. • No fine-tuning because no labeled Sinhala assistive-reader dataset exists. (NLLB-200 (translation) and Llama 3.2:1b (intent extraction) used as pre-trained)
IT22259134	• Need to compare and show performance against other Sinhala spelling and OCR correction algorithms	• Include in the presentation at slide 10, comparing CER;s and WER's of different systems.
IT22120670	• Why is article classification needed? (is it sufficient to just let the LLM know we need the article outcome at the specific level)	• Now it is handled directly through the RAG pipeline: the user's requested intent is passed as part of the prompt to the LLM, along with the retrieved article content.

The Problem and Solution



Problem

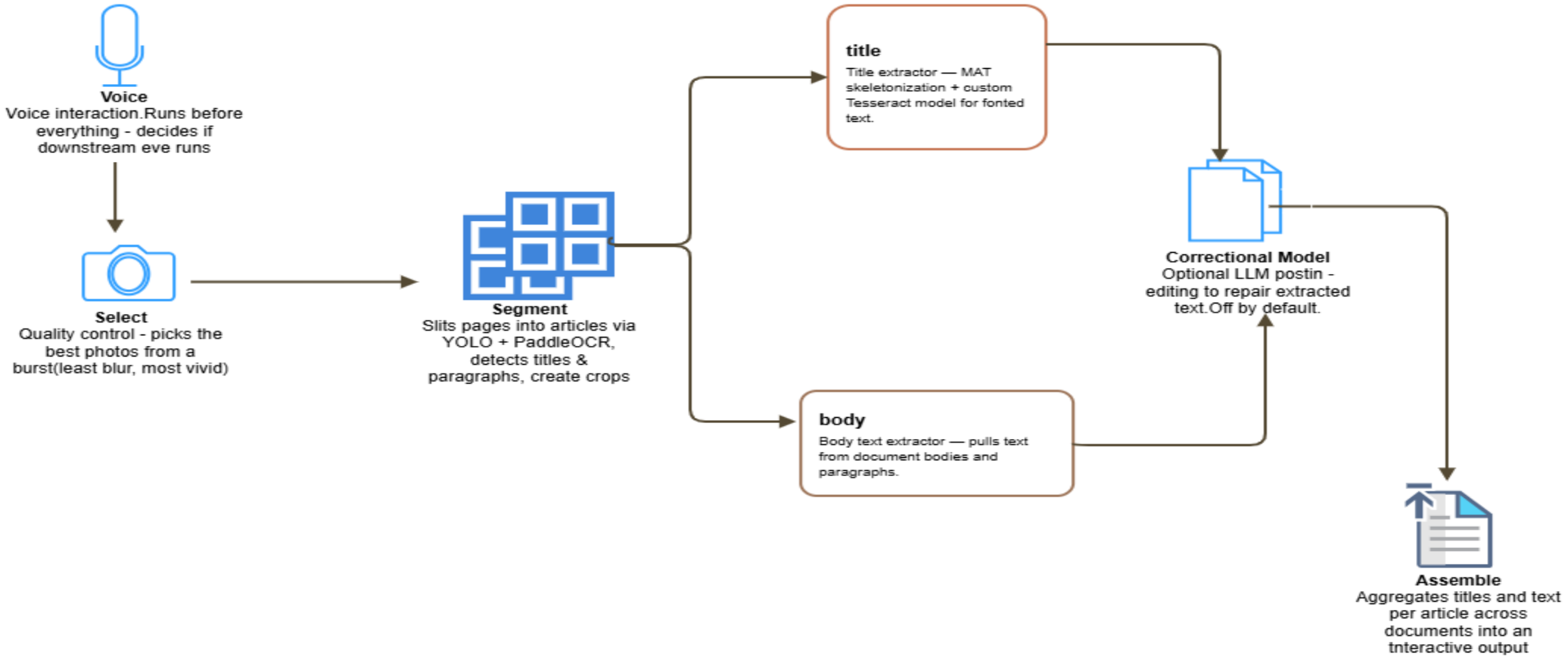
- People with Visual Impairment in Sri Lanka have limited access to information.
- Existing assistive tools like JAWS and Google Lookout are English-centric and offer no functional Sinhala script support.

Users and Stakeholders

- Primary users - visually impaired and elderly Sinhala-speaking individuals who rely on newspapers, textbooks and other documents for daily information.
- Secondary users - OCR developers and research institutions seeking a font-invariant Sinhala recognition, context aware Error correction, conversational Sinhala voice modules for integration.
- Tertiary stakeholders include Sri Lankan disability advocacy groups, government education agencies, and the open-source NLP community.

Visually impaired Sinhala speakers are excluded from printed information due to the absence of affordable, script-aware assistive technology.

System Architecture



IT22072238 | MUTHUNAYAKA M.J.J



Topological Sinhala Character Recognition

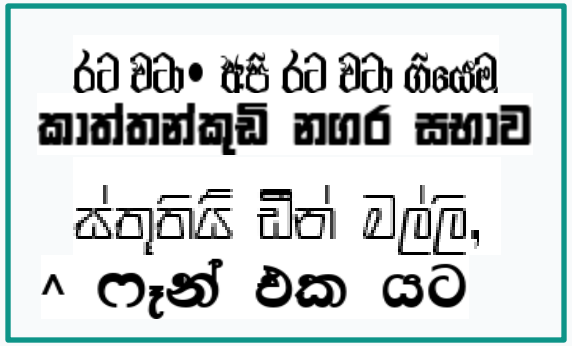
via Medial Axis Transform

Design and Process



[Part 1]

[Joining of Part 1 & 2 by using the below format]



[Part 2]

Extracted text from the images

- Implemented with Paddle OCR to detect text boxes and the CRAFT to properly focus bounds

- Model Used : Tesseract Trained on ~90,000 images of skeletonized images from the SinOCR dataset
- Images were reseized to a height of 384 pixels, skeletonized to a single pixel, then dilated to a total of 3 pixels
- A model trained on regular images is also used side by side as a fallback.

Results



Evaluation Results on Test images of SinOCR

Metric	Regular Model	MAT Model
Baseline CER	0.0586	0.0922
Images won	4,428	1,584
Average Processing time	~213ms	~5,777 ms
Perfect Predictions	524	379
Survey, in which user voted more complex	45%	55%

Evaluation Results on 10k images generated on 135 fonts

Metric	Regular Model	MAT Model
135 fonts stress test – fonts won	105	29
Font Complexity Score	1.25	1.59

Closing Statement

- While RAW remains the stronger and far faster generalist overall, MAT delivers a statistically significant accuracy advantage on the subset of fonts that are objectively most structurally complex, resulting in a valuable, yet computationally expensive, specialist rather than a replacement.

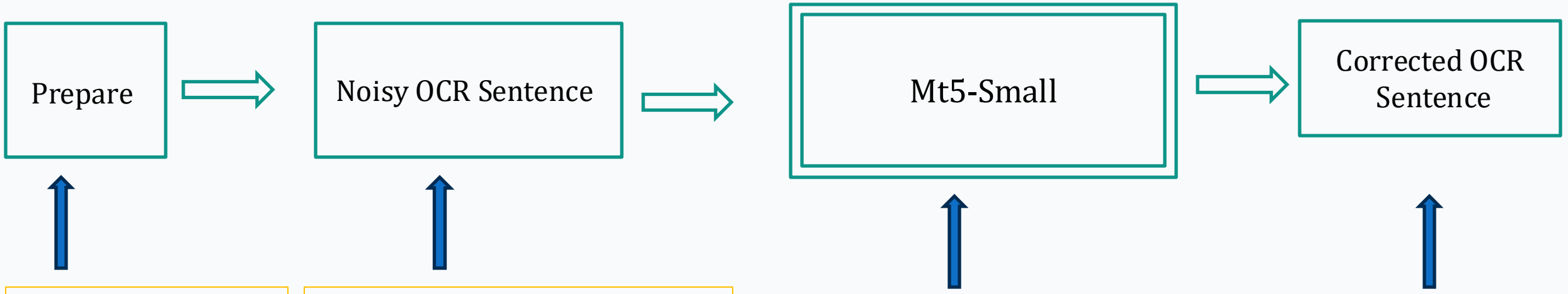
IT22259134 Madhusanka H.P.I



Context-Aware Sinhala Post OCR Error Correction

for Assistive Document Reading

Design and Process



- Crop article regions
- Rescale to target

- Includes errors in categories such as
  - Vowel sign
  - Mixed
  - Noise artifacts
  - Punctuation
  - Hal kirima
  - Independent vowel
  - Space
  - Anusvara.

- Fine-tuned on Sentence pairs of noisy and clean data.
- Sentences made from section extracted from 1232 close-up photographs of individual articles, that where manually corrected.

- Resulting text would on average have a ;
  - +36.7% changes in CER
  - +51.2% change in WER.

Results

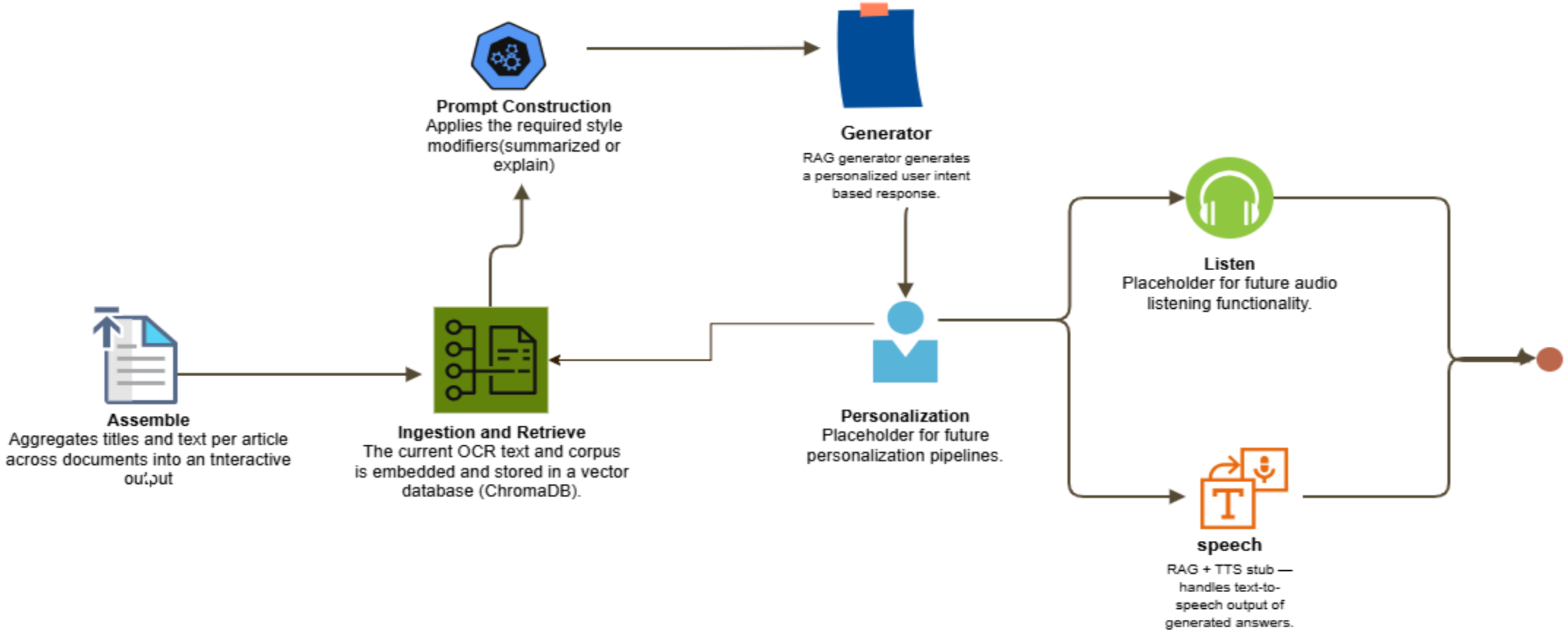
System	CER	WER	% Change CER	% Change WER
Raw OCR (no correction)	0.1197	0.3358	-	-
Dictionary + edit dist	0.1193	0.3187	+0.3%	+5.1%
N-gram spell checker	0.1432	0.3517	-19.7%	-4.7%
Detector-gated	0.1160	0.2938	+3.1%	+12.5%
Full-sequence mt5	**0.0757**	**0.1640**	**+36.7%**	**+51.2%**

Existing vs This Research

System	Context Aware	OCR Specific	Sinhala
Balasooriya 2020	✗	✓	✓
Perera & Sumanthilaka 2025	✓	✗	✓
Guan & Green 2024	✓	✓	✗
This Research	✓	✓	✓

No transformer post-OCR correction exist for Sinhala

System Architecture



IT22120670 | W.G.N DEVINDI

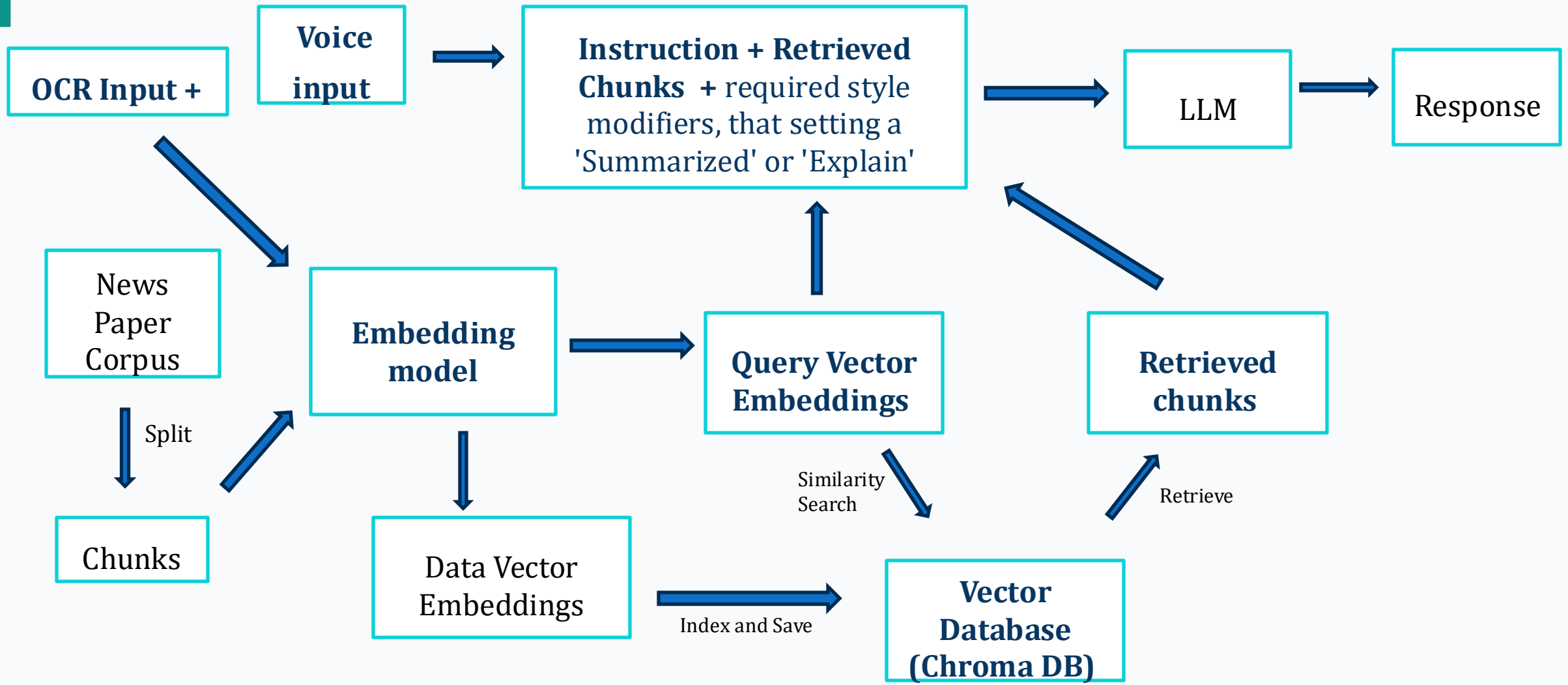


Advanced Document Retrieval and Information Generation

RAG Pipeline for Sinhala Assistive Reading — Component 3

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Design Process



Results

Style	Perplexity Score	Preview
Detailed	1.8263	මෙම පාඨයේ අර්ථය වන්නේ නාමල් රාජපක්ෂ මහතා, පාස්කු දින ප්‍රහාර...
Simple	1.6214	නාමල් රාජපක්ෂ මහතා කියන්නේ මෙයයි. පාස්කු දින ප්‍රහාරයට “රාජප...
Step by Step	1.7271	1. මෙම පාඨය අනුව නාමල් රාජපක්ෂ මහතා කියන්න උත්සාහ කරන්නේ,තා, පාස්කු ප්‍රහාරය...

Table1 - Fluency Variation Across Detailed, Simple, and Step-by-Step Response Styles – To a single corpus – Intent Explain

Style	N	Mean Perplexity Score	Std Dev.
Detailed	25	1.9333	0.0639
Simple	25	1.7857	0.0526
Step by Step	25	1.7899	0.0631

Table2 - Mean Perplexity Scores by Response Style (N=25 Corpus Texts) Intent Explain

Perplexity score measures : The language fluency , grammatical correctness and how natural the generated text sounds

IT22004468 | SATHSARA S.P.Y B

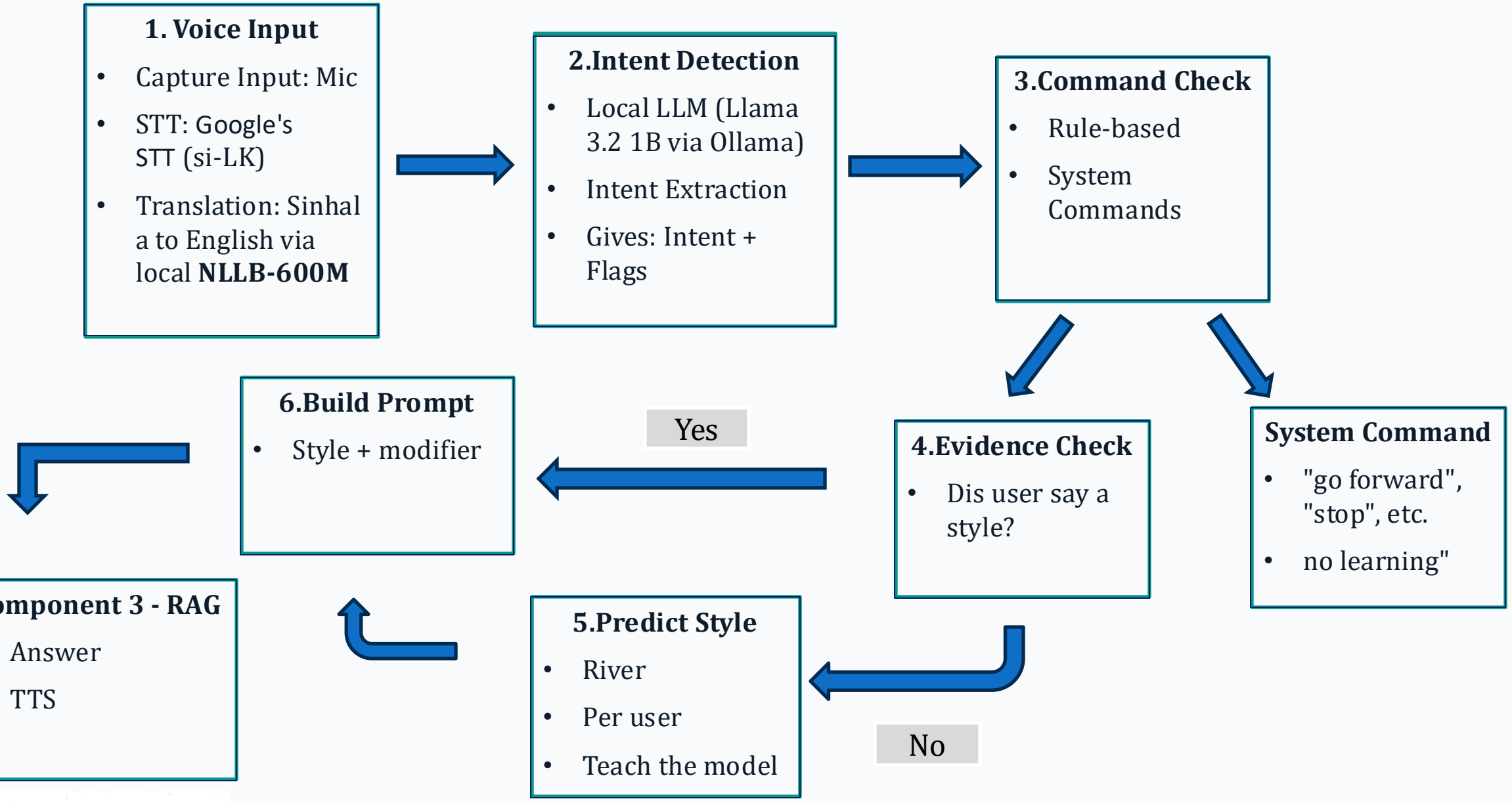


Adaptive Conversational Personalization and Voice Interaction Module

(Personalization and Voice Interaction Module)

Component 4 — Interactive Sinhala Text Reader for the Visually Impaired

Adaptive Conversational Personalization and Voice Interaction Module



Results

Approach	Accuracy	Avg Latency
Approach 1 — NLLB + Llama 3.2:1b	85.7%	7.38s
Approach 2 — Qwen 2.5:1.5b (direct)	0.8%	8.21s
Approach 3 — Trained (TF-IDF + LinearSVC)	57.1%	1.42s
Hybrid (trained model + LLM fallback)	71.4%	2.79s

🏆 Best accuracy: **Approach 1**

User Voice Input	Evidence Source	User Profile History	Results
"What is this document?"	None (Neutral question)	Simple: 0.0 | Detailed: 0.0 | Step: 0.0 (n_confirmed = 0)	Detailed (Default)
"Please explain this simply	Explicit Intent (SIMPLIFY)	Simple: 1.0 | Detailed: 0.0 | Step: 0.0 (n_confirmed = 1)	Simple (Model Learns)
"What is written in section 2?"	None (Neutral question)	Simple: 89% | Detailed: 11% | Step: 0% (Normalized weights)	Simple (Model Predicted)
"No, give me the full details"	Correction Detected	Simple: 45% | Detailed: 55% | Step: 0% (Decayed old weights)	Detailed (Model Adapts)

Parameter	Value
Algorithm	Softmax Regression (River)
Learning Mode	Online (incremental, per-turn)
Feature Set	TF-IDF (text) + User history weights
Output Classes	Simple, Detailed, StepByStep
History Decay Factor	0.85 (exponential recency weighting)

Thank You!

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